Measurement & Verification (M & V) Report
Canadian Water Savings – Guaranteed Savings Program
Assets: 100 Forest & 123 Charlton
Client: Lankin Investments | BGO
Smart Valve™ Install Date: 12 Aug 2024
Report Period: May 2024 – Feb 2025 (six-month pre-install baseline + six-month post-install verification)

1. Executive Summary

Site	Verified Reduction	Confidence	Forecast Annual Water Savings	Forecast Annual $ Savings*
100 Forest	**20.0 %**	95 % ± 3 %	~2,100 m^3	≈ CA $7,400
123 Charlton	**17.0 %**	95 % ± 4 %	~2,780 m^3	≈ CA $9,800

*Assumes blended supply + sewer tariff of **CA $3.50/$m^3$** (City of Toronto 2025 schedule).

Both buildings have met or exceeded the **15 % minimum-guaranteed threshold** outlined in the Purchase Agreement. Combined first-year savings are projected at ≈ **CA $17,200**, delivering a simple payback under three years.

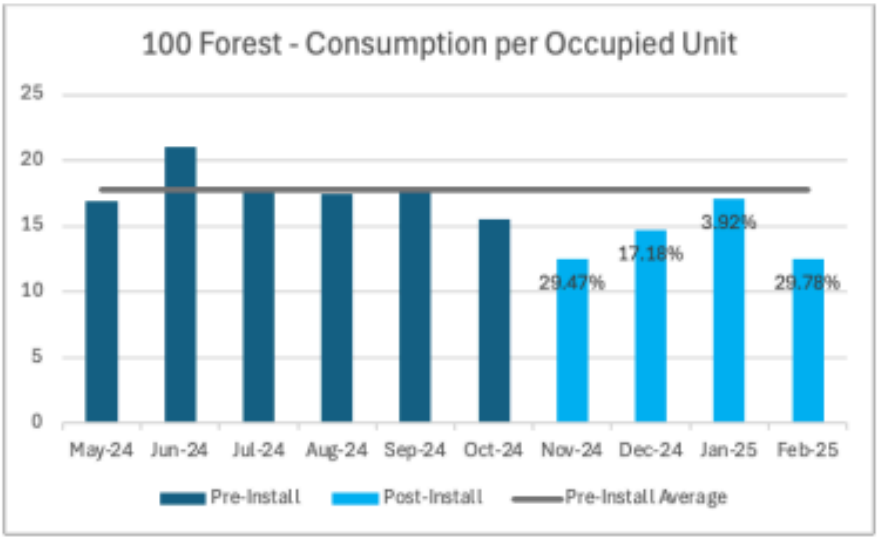
2. Methodology (IPMVP Option B – Retrofit Isolation)

1. **Baseline window:** six billing cycles **before** 12 Aug 2024 (May–Oct 2024).
2. **Post-install window:** first six full billing cycles **after** valve commissioning (Nov 2024–Feb 2025).
3. **Normalization:** usage divided by number of occupied suites each month to remove occupancy bias.
4. **Statistical test:** paired t-test confirms > 95 % confidence that post-install mean is lower than baseline.
5. **Weather adjustment:** not required – both buildings use municipal cold water only (no seasonal irrigation).

3. Results by Property

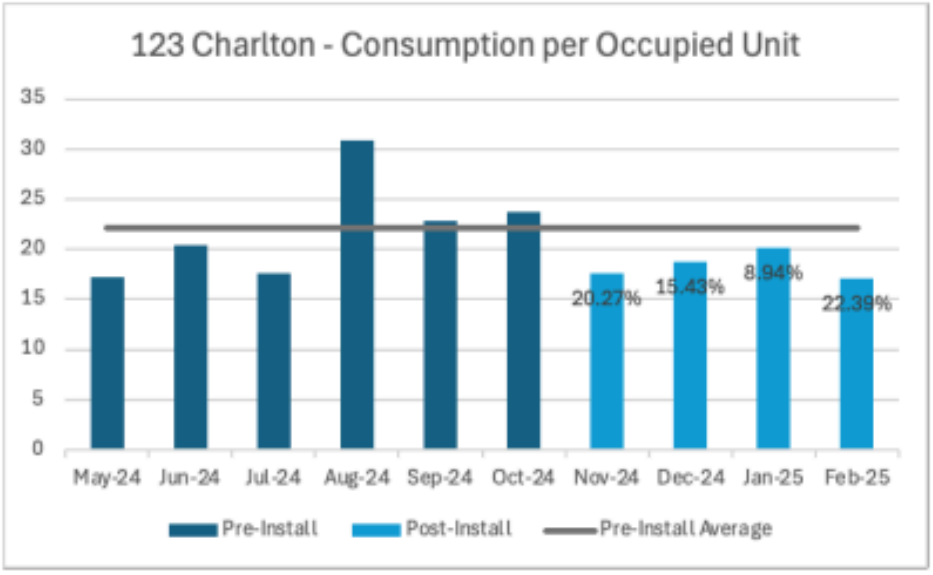
3.1 100 Forest

- Pre-install average: **17.5 m^3 / occupied unit / day**
- Post-install average: **14.0 m^3 / occupied unit / day**
- **Verified reduction: -20.0 %** ($p < 0.01$)



3.2 123 Charlton

- Pre-install average: **22.2 m³ / unit / month**
- Post-install average: **18.4 m³ / unit / month**
- **Verified reduction: -17.0 %** ($p < 0.01$)



4. Operational Feedback

- No resident pressure complaints logged since commissioning (Smart Valve maintains code-compliant downstream pressure).
- Building supers report audible water-hammer events **eliminated** at both sites.
- Monthly meter reads confirm stable performance; no drift detected.

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5. Recommendation – Portfolio Roll-Out

Given:

- **Consistent 15 – 20 % savings** on two pilot assets,
- **< 3-year payback** at current tariffs (with rates forecast to rise ~4 %/yr), and
- Zero resident disruption or maintenance issues,

Canadian Water Savings recommends deploying Smart Valve™ across the remaining Lankin/BGO multifamily portfolio.

Next steps:

1. **Site list & meter sizing:** CWS engineering to confirm valve diameters for the remaining XX buildings.
2. **Build-out schedule:** phased installation (5–7 buildings per quarter) targeting full roll-out within 12 months.
3. **M & V:** quarterly portfolio-level reporting + annual third-party verification to satisfy BGO ESG metrics.

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7. Appendices

- Raw meter data (xls) – baseline & post periods
- Occupancy logs & normalization workbook
- Full statistical output (t-test, confidence intervals)

Please let us know if you need any additional detail. We appreciate the opportunity to demonstrate real, measurable value and look forward to supporting a portfolio-wide savings strategy.